COURSE NAME

SOFTWARE ENGINEERING

CSC 3114

(UNDERGRADUATE)

CHAPTER 6

THE DYNAMIC SYSTEMS DEVELOPMENT METHOD (DSDM)

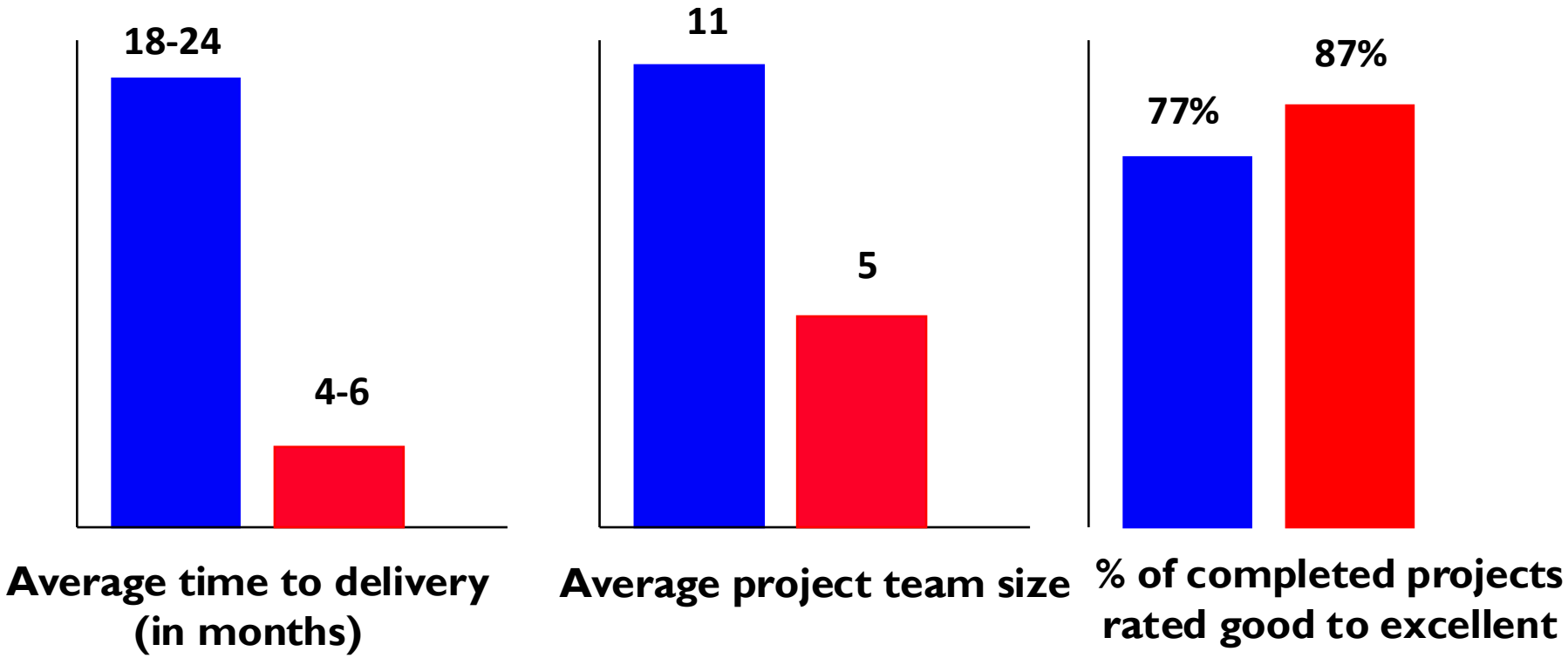
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DSDM

- **The Dynamic Systems Development Method (DSDM)** is a public domain Rapid Application Development method which has been developed through capturing the experience of a large group of vendor and user organisations.
- **It is now considered to be the UK's de-facto standard for RAD.**

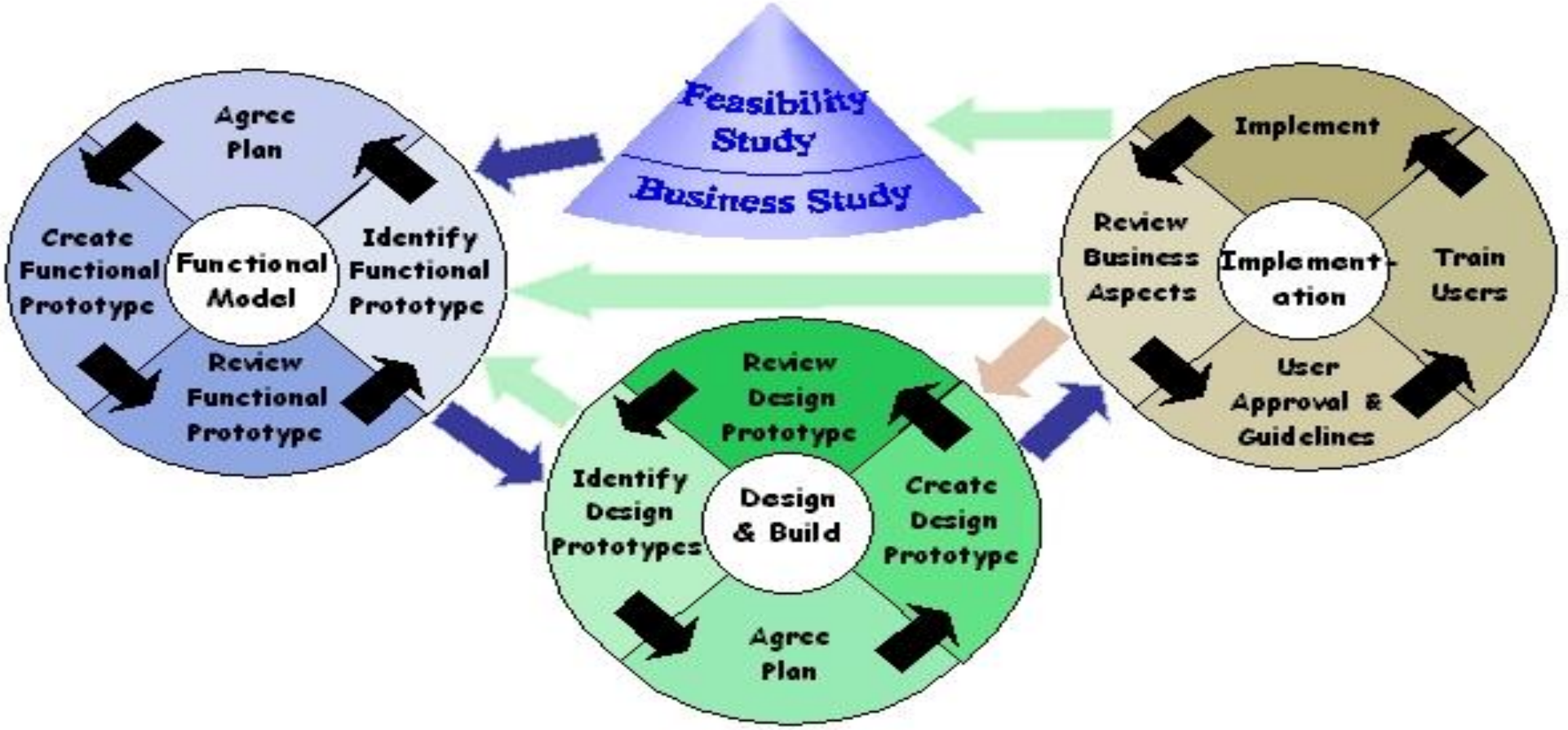
- The key to DSDM is to deliver **what** business needs **when** it needs.
  - Achieved by using the various techniques in the framework and flexing requirements
  - The aim is always to address the current and imminent needs of the business rather than to attack all the perceived possibilities

TRADITIONAL METHOD VS. DSDM



Source: British Airways IM Department, Newcastle

DSDM PROCESS VIEW



DSDM PROCESS

Activity	Sub activity	Description
Study	**Feasibility Study**	Stage where the suitability of DSDM is assessed. Judging by the type of project, organizational and people issues, the decision is made, whether to use DSDM or not. Therefore it will generate a FEASIBILITY REPORT, a FEASIBILITY PROTOTYPE, and a GLOBAL OUTLINE PLAN which includes a DEVELOPMENT PLAN and a RISK LOG.
	Business Study	Stage where the essential characteristics of business and technology are analyzed. Approach to organize workshops, where a sufficient number of the customer's experts are gathered to be able to consider all relevant facts of the system, and to be able to agree on development priorities. In this stage, a PRIORITIZED REQUIREMENTS LIST, a BUSINESS AREA DEFINITION, a SYSTEM ARCHITECTURE DEFINITION, and an OUTLINE PROTOTYPING PLAN are developed.

DSDM PROCESS

Activity	Sub activity	Description
Functional Model Iteration	Identify functional prototype	Determine the functionalities to be implemented in the prototype that results from this iteration. In this sub-stage, a FUNCTIONAL MODEL is developed according to the deliverables result of business study stage.
	Agree schedule	Agree on how and when to develop these functionalities.
	Create functional prototype	Develop the FUNCTIONAL PROTOTYPE, according to the agreed schedule and FUNCTIONAL MODEL.
	Review functional prototype	Check correctness of the developed prototype. This can be done via testing by end-user and/or reviewing documentation. The deliverable is a FUNCTIONAL PROTOTYPING REVIEW DOCUMENT.

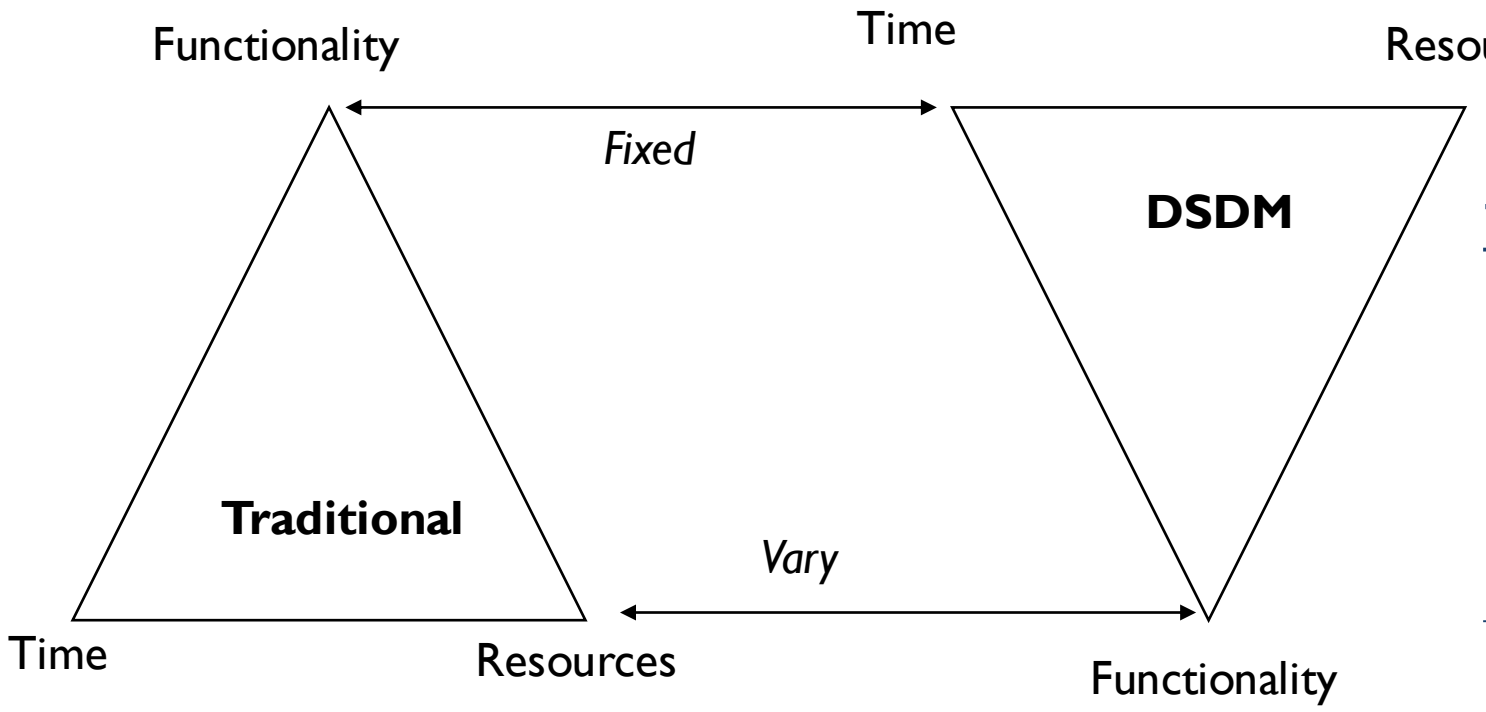
DSDM PROCESS

Activity	Sub activity	Description
Design and Build Iteration	Identify design prototype	Identify functional and **non-functiona**l requirements that need to be in the tested system. And based on these identifications, an IMPLEMENTATION STRATEGY is involved. If there is a TEST RECORD from the previous iteration, then it will be also used to determine the IMPLEMENTATION STRATEGY.
	Agree schedule	Agree on how and when to realize these requirements.
	Create design prototype	Create a system (DESIGN PROTOTYPE) that can safely be handed to end-users for daily use, also for testing purposes.
	Review design prototype	Check the correctness of the designed system. Again, testing and reviewing are the main techniques used. A USER DOCUMENTATION and a TEST RECORD will be developed.

DSDM PROCESS

Activity	Sub activity	Description
Implementation	User approval and guidelines	End users approve the tested system (APPROVAL) for implementation and guidelines with respect to the implementation and use of the system are created.
	Train users	Train future end user in the use of the system. TRAINED USER POPULATION is the deliverable of this sub-stage.
	Implement	Implement the tested system at the location of the end users, called as DELIVERED SYSTEM.
	Review business	Review the impact of the implemented system on the business, a central issue will be whether the system meets the goals set at the beginning of the project. Depending on this the project goes to the next stage, the post-project or loops back to one of the preceding stages for further development. This review is will be documented in a PROJECT REVIEW DOCUMENT.

DIFFERENCE BETWEEN TRADITIONAL DEVELOPMENT VS. DSDM



Traditional Method

- Functional/requirements are fixed
- Time & resources can vary

DSDM/Agile Methods

- Functional/requirement varies
- Time & resources are fixed

TECHNIQUES TO CONSIDER IN DSDM

- Flexibility [80:20 Rule]
- Timeboxing
- MoSCoW Rules
- Prototyping

DSDM TECHNIQUES: FLEXIBILITY

- A fundamental assumption of DSDM is that nothing is built perfectly first time.
- 80:20 Rule: The 80/20 rule says that you can get 80% of the value from a project by focusing on the 20% of needs that are the most important.

DSDM assumes that all previous steps can be revisited as part of its iterative approach.

Therefore, the current step need be completed only enough to move to the next step, since it can be finished in a later iteration.

DSDM TECHNIQUES: TIMEBOXING

- Without effective timeboxing, prototyping teams can lose their focus and run out of control.
- Timeboxing works by concentrating on when a business objective will be met as opposed to the tasks which contribute to its delivery.
- **Timeboxing Basics**
  - Time between start and end of an activity
  - DSDM uses nested timeboxes, giving a series of fixed deadlines
  - Ideally **2 - 4 weeks** in length
  - Objective is to have easiest 80% produced in each timebox
  - Remaining 20% potentially carried forward subsequent timeboxes
  - Focus on the essentials
  - Helps in estimating and providing resources

DSDM TECHNIQUES: MOSCOW RULES

- **MoSCoW** rules formalised in DSDM version 3
  - **Must have** – fundamental to project success
  - **Should have** – important but project does not rely on
  - **Could have** – left out without impacting on project
  - **Want to have but Won't have** this time for those valuable requirements that can wait till later development takes place; in other words, the Waiting List.

Category	Meaning	Example
Must-have	**Essential requirements** that are critical for system success.	User login functionality; data encryption; payment processing.
Should-have	**Important features**, but not vital for initial delivery.	Password reset option; mobile responsiveness; role-based access.
Could-have	**Nice-to-have enhancements** that have little impact if left out.	Dark mode; advanced analytics dashboard; chatbot.
Won't-have (this time)	**Least priority**, out of scope for now, but may be considered in future.	Third-party integrations; multilingual support; gamification.

DSDM TECHNIQUES: PROTOTYPING

Prototypes are necessary in DSDM because

- Facilitated workshops define the high-level requirements and strategy
- Prototypes provide the mechanism through which users can ensure that the detail of the requirements is correct
- Demonstration of a prototype broadens the users awareness of the possibilities and assists them in giving feedback to the developers
- Speeds up the development process and increases confidence that the right solution will be delivered

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